

Lecture 8: The Ising model

Introduction

- ▶ Up to now: Toy systems with interesting properties (random walkers, cluster growth, percolation)
- ▶ Common to them: No interactions
- ▶ Add interactions now, with significant role
- ▶ Immediate consequence: much richer structure in model, in particular: phase transitions
- ▶ Simulate interactions with RNG ([Monte Carlo method](#))
- ▶ Include the impact of temperature: ideas from thermodynamics and statistical mechanics important
- ▶ Simple system as example: coupled spins (see below), will use the canonical ensemble for its description

The Ising model

- ▶ A very interesting model for understanding some properties of magnetic materials, especially the phase transition ferromagnetic $\longleftrightarrow$ paramagnetic
- ▶ Intrinsically, magnetism is a quantum effect, triggered by the spins of particles aligning with each other
- ▶ Ising model a superb toy model to understand this dynamics
- ▶ Has been invented in the 1920's by E.Ising
- ▶ Ever since treated as a first, paradigmatic model

The model (in 2 dimensions)

- ▶ Consider a square lattice with spins at each lattice site
- ▶ Spins can have two values: $s_i = \pm 1$
- ▶ Take into account only nearest neighbour interactions (good approximation, dipole strength falls off as $1/r^3$)
- ▶ Energy of the system:

$$E = -J \sum_{\langle ij \rangle} s_i s_j$$

- ▶ Here: exchange constant $J > 0$ (for ferromagnets), and $\langle ij \rangle$ denotes pairs of nearest neighbours.
- ▶ (Micro-)states α characterised by the configuration of each spin, the more aligned the spins in a state α , the smaller the respective energy E_α .
- ▶ Emergence of **spontaneous magnetisation** (without external field): sufficiently many spins parallel

Adding temperature

- ▶ Without temperature T : Story over.
- ▶ With temperature: disordering effects, spins flip randomly.
- ▶ In the following assume system to be in contact with external heat bath - can use canonical ensemble (look it up, if you're interested).
- ▶ Effect: System will again “explore” all possibilities (like in the case of milk in tea, lecture 6)
- ▶ In contrast here: No uniform probability, but

$$\mathcal{P}_\alpha \sim \exp \left[-\frac{E_\alpha}{k_B T} \right] \text{ (Boltzmann factor)}$$

for the system to be in a micro-state α .

- ▶ Consequence: Increasing temperatures open micro-states with larger energies $\longleftrightarrow$ less alignment

Observables

- ▶ How to calculate macroscopic observables?
Example: the magnetisation of the system M .
- ▶ First step: calculate the observable for a micro-state:

$$M_{\alpha} = \sum_i s_i$$

- ▶ Sample over all micro-states:

$$M = \sum_{\alpha} M_{\alpha} \mathcal{P}_{\alpha}$$

- ▶ Therefore, name of the game:
How to calculate the $\mathcal{P}_{\alpha}$ /sample over the micro-states.
- ▶ Will present two solutions in the following

Analytic solution: Mean field theory (MFA)

- ▶ Mean field theory an extremely powerful **approximation**
- ▶ Will introduce it with the specific example of the Ising model at hand.
- ▶ However nice: It is **not precise!**
- ▶ Basic idea: Replace the individual spins $s_i = \pm 1$ with an average value $\langle s \rangle \in [-1, 1]$. Then

$$M = \sum_i s_i \longrightarrow \sum_i \langle s \rangle = N \langle s \rangle \equiv N \langle s_i \rangle$$

- ▶ In other words: Deduce an **average alignment** (works for an infinitely large system - all spins equivalent).
- ▶ With the equation above we used that we can pick every spin as average spin. But how can we calculate without the micro-states?
- ▶ Answer: A trick (see below)

The trick for the solution

- ▶ Add a magnetic field (seems a detour, but wait & see!):

$$E = -J \sum_{\langle ij \rangle} s_i s_j - \mu H \sum_i s_i$$

(Magnetic field H interacts with spins through their magnetic moment μ .)

- ▶ Consider a system made of one **single spin**: $E_{\pm} = \mp \mu H$.

- ▶ Two micro-states with $\mathcal{P}_{\pm} = C \exp \left[\pm \frac{\mu H}{k_B T} \right]$

- ▶ Normalisation C from $\mathcal{P}_+ + \mathcal{P}_- = 1$

$$\implies C = \frac{1}{\exp\left(\frac{\mu H}{k_B T}\right) + \exp\left(-\frac{\mu H}{k_B T}\right)} = \frac{1}{2 \cosh\left(\frac{\mu H}{k_B T}\right)}$$

- ▶ Therefore thermal average of the single spin:

$$\langle s_i \rangle = \mathcal{P}_+ - \mathcal{P}_- = \tanh \frac{\mu H}{k_B T}$$

Rounding it off

- ▶ Having the solution for a single spin in a background field, we **replace the background field with the average spins!**

$$E = - \sum_i \left(\sum_{\langle ij \rangle} s_j + \mu H \right) s_i = -\mu H_{\text{eff}} \sum_i s_i$$

- ▶ The effective magnetic field is thus

$$H_{\text{eff}} = \frac{J}{\mu} \sum_{\langle ij \rangle} s_j + H$$

- ▶ Mean field approximation then sets $H = 0$ and replaces the actual s_j with the average value:

$$H_{\text{mfa}} = \frac{jJ}{\mu} \langle s \rangle$$

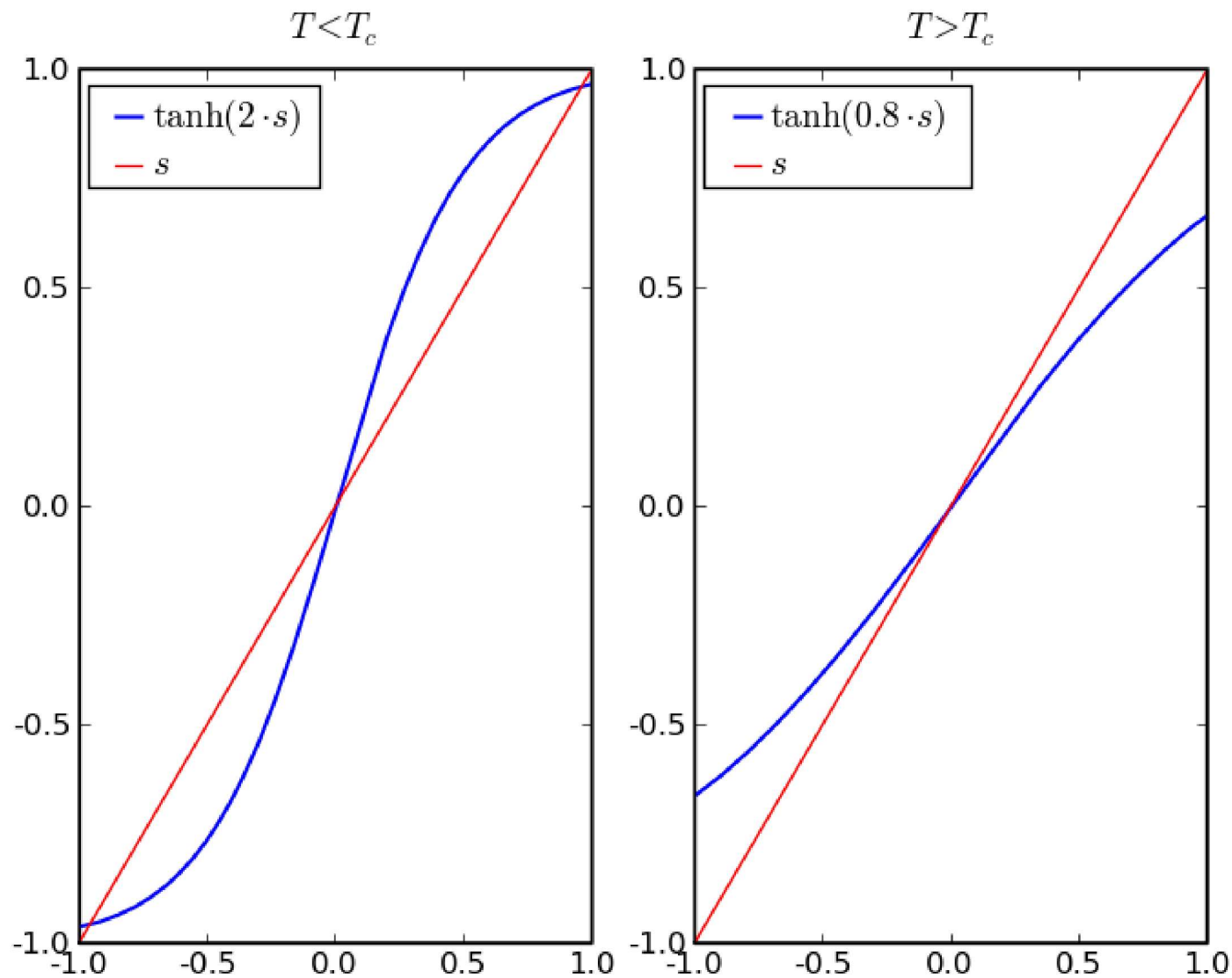
(Here $j = 4$ is number of nearest neighbours)

- ▶ This implies the following implicit equation:

$$\langle s \rangle = \tanh \frac{jJ \langle s \rangle}{k_B T}$$

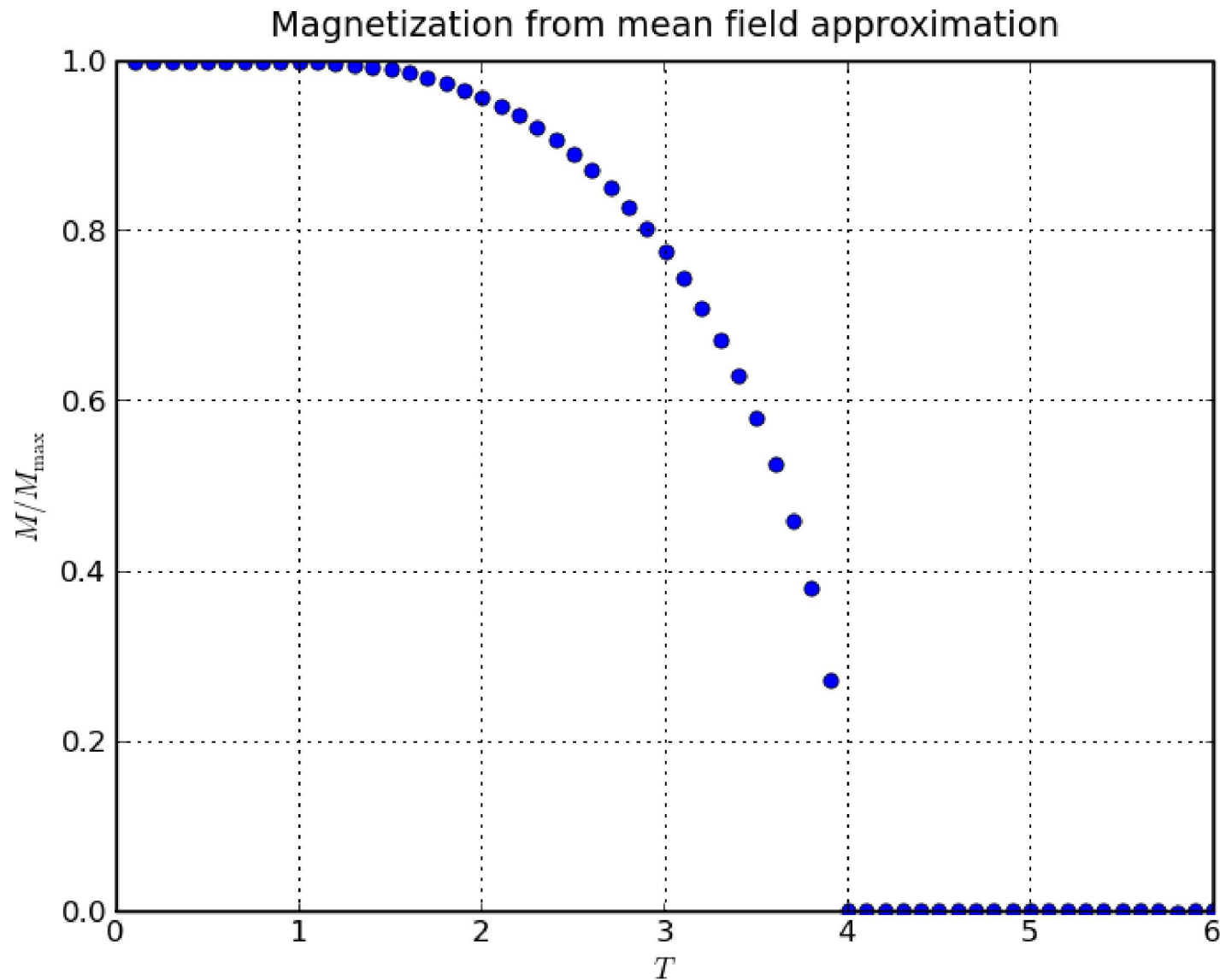
Results: Schematic

- Notice two different regimes: either 1 or 3 solutions



True results

- Obtained from solving $f(\langle s \rangle) = \langle s \rangle - \tanh \frac{4\langle s \rangle}{T} = 0$



Discussion of results

- ▶ In plots use $M_{\max} = N$ for normalisation and $J = k_B$ to keep things simple.
- ▶ Phase transition at $T_c = 4$ - second order
(1st derivative of **order parameter magnetisation** jumps)
- ▶ Around T_c : $dM/dT \rightarrow \infty$.
- ▶ Exact form of singularity from Taylor expansion of $\tanh$:
$$\tanh x = x - \frac{x^3}{3} + \mathcal{O}(x^4)$$
- ▶ Therefore, around $T = T_c$:

$$\langle s \rangle = \frac{jJ}{k_B T} \langle s \rangle - \frac{1}{3} \left(\frac{jJ}{k_B T} \right)^3 \langle s \rangle^3$$

Quantifying the phase transition

- ▶ Non-trivial solution for:

$$\langle s \rangle = \sqrt{\frac{3}{T} \left(\frac{k_B T}{jJ} \right)^3 \left(\frac{jJ}{k_B} - T \right)^{1/2}} \sim (T_c - T)^\beta$$

- ▶ Critical temperature and exponent:

$$T_c = jJ/k_B, \beta = 1/2.$$

- ▶ But exact results (analytically known):

$$T_c = 2/\ln(1 + \sqrt{2}) \approx 2.27, \beta = 1/8$$

for a square lattice with $j = 4$ and $J = k_B$.

- ▶ Will now turn numerical/simulation

Strategy of simulation

- ▶ Strategy very similar to what's been done before: Use the random number generator to evolve the system
- ▶ This time: RNG to describe interactions
- ▶ Necessary: Interactions in probabilistic language
- ▶ Algorithm will look like: Go over the spins, check whether they flip (compare $\mathcal{P}_{\text{flip}}$ with number from RNG), repeat to equilibrate.
- ▶ To calculate $\mathcal{P}_{\text{flip}}$: Use energy of the two micro-states (before and after flip) and Boltzmann factors.
- ▶ While running, evaluate observables directly and take thermal average (average over many steps).

Metropolis

A classical method for the program above

1. Initialise the lattice, i.e. fix all s_i
(either at random, or $s_i = 1 \forall i$, or similar)
2. In each time step go through the entire lattice.
For each spin decide whether it flips or not
 - ▶ Calculate $E = -J \sum s_i s_j$ for both configs
 - ▶ Determine characteristic “energy gain by flip”
 $E_{\text{flip}} = E_{\text{after}} - E_{\text{before}}$
 - ▶ if $E_{\text{flip}} < 0$, the spin is flipped
 - ▶ if $E_{\text{flip}} > 0$, the spin is flipped with probability
$$\mathcal{P}_{\text{flip}} = \exp\left(-\frac{E_{\text{flip}}}{k_B T}\right)$$
3. Repeat with a huge number of time steps
 $\implies$ allow the system to equilibrate

Why Metropolis is correct: Detailed balance

- ▶ Consider one spin flip, connecting micro-states 1 and 2.
- ▶ Rate of transitions given by the transition probabilities $\mathcal{W}$
- ▶ If $E_1 > E_2$ then $\mathcal{W}_{1 \rightarrow 2} = 1$ and $\mathcal{W}_{2 \rightarrow 1} = \exp\left(-\frac{E_1 - E_2}{k_B T}\right)$
- ▶ In thermal equilibrium, both transitions equally often:

$$\mathcal{P}_2 \mathcal{W}_{2 \rightarrow 1} = \mathcal{P}_1 \mathcal{W}_{1 \rightarrow 2}$$

This takes into account that the respective states are occupied according to the Boltzmann factors.

- ▶ In principle, all systems in thermal equilibrium can be studied with Metropolis - just need to write transition probabilities in accordance with detailed balance, as above.
- ▶ Metropolis algorithm simulates the canonical ensemble, summing over micro-states with a Monte Carlo method.

Some code specifics

Initialise the $L \times L$ lattice with spins s_i .

Take the spins either fixed (all $s_i = 1$) or at random (all $s_i = \pm 1$)

A single time-step: sweep through the lattice

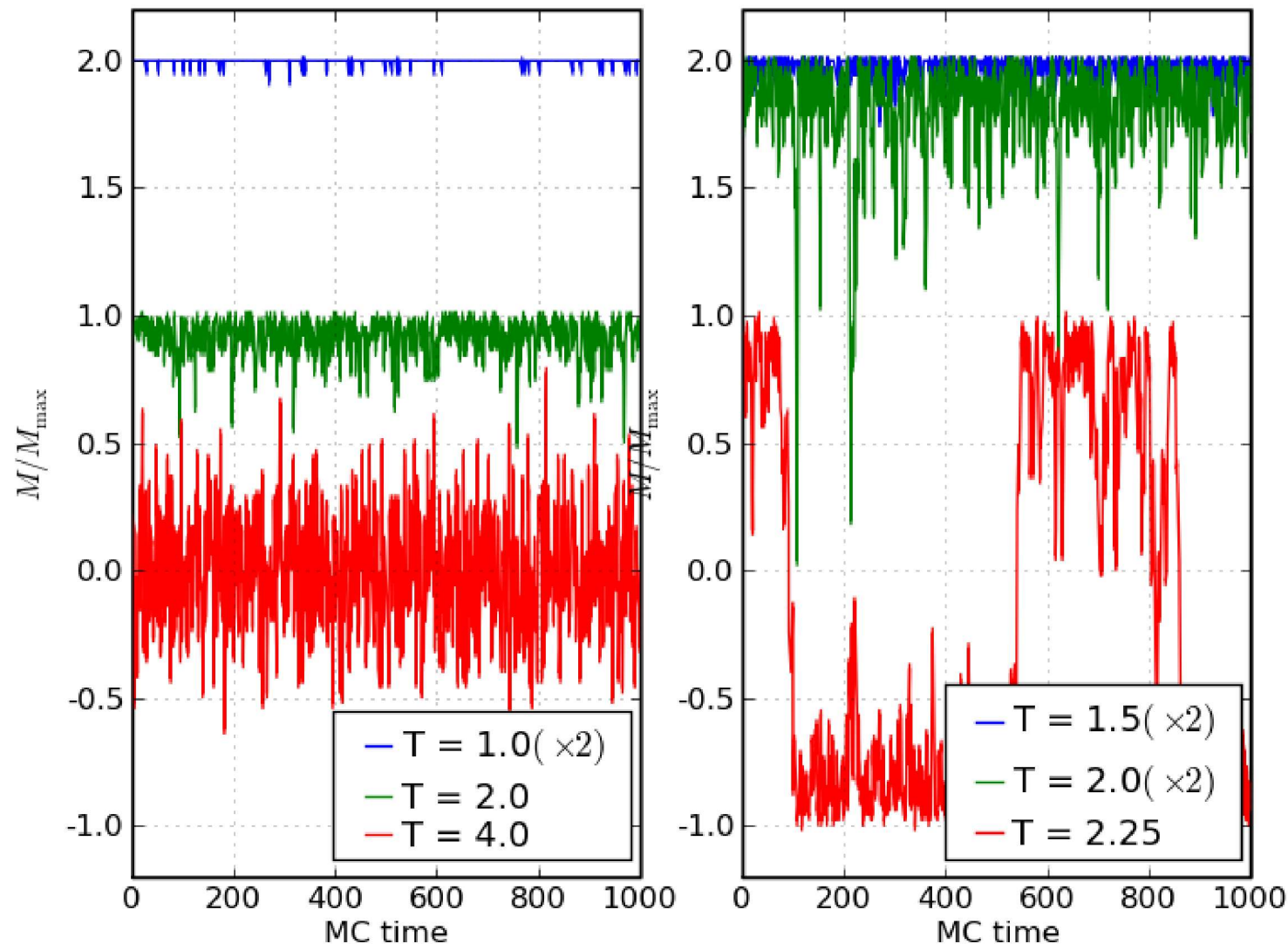
Go systematically through the lattice, line by line, spin by spin, and decide, whether the spin should flip or not. Important here: boundary conditions

Boundary conditions (finite-size lattices)

We use **periodic boundary conditions**. This means that the spins at the right edge of the lattice are taken as neighbours of spins at the left edge. For 2-D lattices this renders our lattice effectively a donut-shaped object. Such a treatment reduces finite-size effects, but one should keep in mind that correlations with a length larger than $\sqrt{2}L$ cannot be simulated.

Sweeping through the lattice

- Fix temperature, use a 10×10 lattice

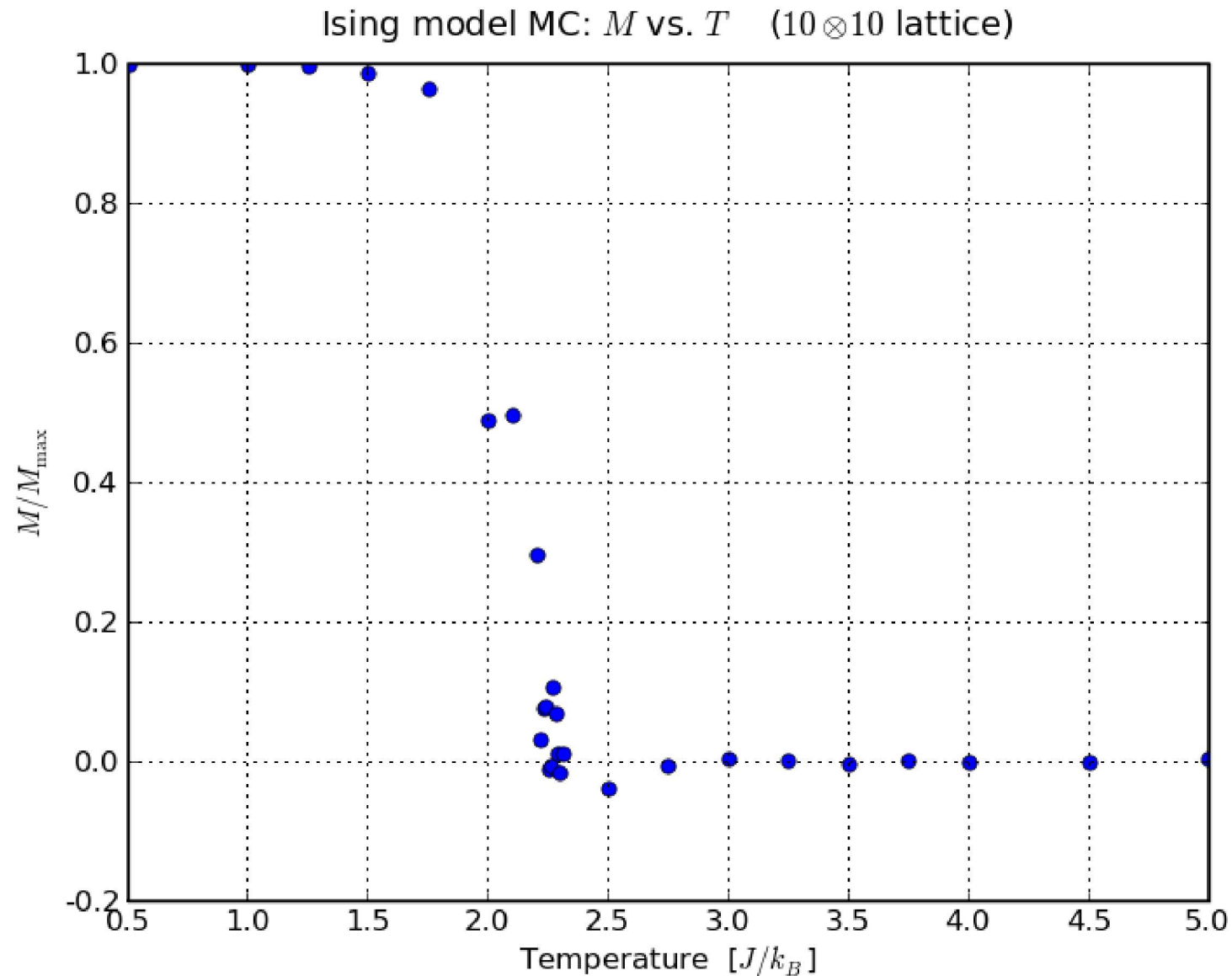


Discussion of the sweeps

- ▶ At low temperatures ($T = 1$): system quite stable, only small fluctuations, relative magnetisation around 1.
- ▶ At larger temperatures, but below T_c ($T=2$): larger fluctuations due to more favourable Boltzmann factor (10% anti-aligned), average magnetisation around 0.9.
- ▶ At large temperatures ($T = 4$): system disordered, no average magnetisation left.
- ▶ Around the **critical temperature** ($T = 2.25$): Huge fluctuations, after periods of stability, system jumps from sizable positive to negative magnetisation and vice versa.
- ▶ This is in agreement with **fluctuation-dissipation theorem** (next lecture).

Phase transition - the MC look at things

- Still on a 10×10 lattice



Some simulation nitty-gritty

- ▶ Results above after equilibrating the system, but
- ▶ **critical slowdown** around critical point:
The system's time to equilibrate diverges.
- ▶ Independent of this: Monte Carlo results in agreement with exact calculation and in disagreement with the results from mean field approximation.

Summary

- ▶ First simulation of a system with interactions
- ▶ Used the Ising model as laboratory: well-defined, well-studied system, analytical results known, a favourite of the simulators
- ▶ A (simple) analytical approximation: mean field theory gives **qualitatively correct results**: existence of a phase transition, estimate of critical temperature
- ▶ Exact calculations (and simulation) agree and are **quantitatively different** from MFA.